

Religion, Social Cohesion, and Economic Structure in Predicting Crime Rate

One-Page Summary (Week 4 Regression-Based Reframing)

Background & Shift from Week 3

Week 3's chi-square test on religion vs. non-religion found no direct effect on crime ($\chi^2 = 1.5054$, $p = 0.9591$). Guided by feedback, Week 4 reframes the question structurally: Can regional economic capacity and social cohesion jointly explain variation in crime rates?

Model & Variables

- Multiple linear regression: $\text{Crime Rate} = \beta_0 + \beta_1(\text{Fiscal Independence}) + \beta_2(\text{Religion Ratio}) + \beta_3(\text{Volunteer Rate}) + \epsilon$.
- Expectations: all predictors have negative signs (higher economic autonomy and cohesion $\rightarrow$ lower crime).
- Interpretation goal: quantify strength (betas) and fit (R^2), not just significance.

Key Results

- Simple regressions (y on each x): Fiscal Independence $\beta \approx -0.152$ ($R^2 \approx 0.13$), Religion Ratio $\beta \approx -0.529$ ($R^2 \approx 0.77$), Volunteer Rate $\beta \approx -1.395$ ($R^2 \approx 0.85$).
- Multiple regression: $\beta(\text{Fiscal}) \approx -0.085$ ($p = 0.015$), $\beta(\text{Religion}) \approx -0.042$ ($p = 0.032$), $\beta(\text{Volunteer}) \approx -0.098$ ($p = 0.006$); Model $R^2 \approx 0.73$.
- Implication: Social participation is the strongest predictor; religion's effect appears through civic/communal pathways.

Integration with Week 3

- Null direct effect in Week 3 becomes a meaningful indirect effect once structural and social variables are included.
- Transition from binary significance testing to explanatory prediction with controls (confounders).

Limitations

- Multicollinearity between Religion Ratio and Volunteer Rate ($r \approx 0.9$): consider PCA or a composite social-capital index.
- Ecological fallacy: regional relationships do not imply individual behavior.
- Cross-sectional design: limits causal claims.
- Data are reconstructed (Seoul KOSIS/Survey inspired) for illustration-not official statistics.

Takeaways & Decision Frame

- $\beta(\text{Volunteer}), \beta(\text{Religion}) < 0$ with $p < 0.05 \rightarrow$ strong support for the indirect-effect hypothesis via social cohesion.
- $\beta(\text{Fiscal}) < 0$ (modest) $\rightarrow$ economic autonomy helps but is insufficient without community engagement.
- High fit ($R^2 \approx 0.73$) $\rightarrow$ structural model captures much of the variance in regional crime rates.

Collaboration (Condensed)

Model redesign: Lim; Theory: Kim; Empirics: Lee; Visualization: Song; Synthesis/Writing: Chung.

Conclusion

Religion influences morality indirectly by building networks of trust and civic participation. Volunteerism and religious participation jointly suppress regional crime; fiscal independence adds a weaker, consistent effect. Morality and crime are emergent outcomes of system-level interactions, not isolated beliefs.